Here Δq_i is the i[th] charge element, r_{iP} is the distance of the point P from the i[th] charge element and $\hat{r}_{iP}$ is the unit vector from i[th] charge element to the point P.

However the equation (1.9) is only an approximation. To incorporate the continuous distribution of charge, we take the limit $\Delta q \to 0 (= dq)$. In this limit, the summation in the equation (1.9) becomes an integration and takes the following form

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r^2}\hat{r} \qquad (1.10)$$

Here r is the distance of the point P from the infinitesimal charge dq and $\hat{r}$ is the unit vector from dq to point P. Even though the electric field for a continuous charge distribution is difficult to evaluate, the force experienced by some test charge q in this electric field is still given by $\vec{F} = q\vec{E}$.

(a) If the charge Q is uniformly distributed along the wire of length L, then linear charge density (charge per unit length) is $\lambda = \frac{Q}{L}$. Its unit is coulomb per meter (Cm^{-1}).

The charge present in the infinitesimal length dl is $dq = \lambda dl$. This is shown in Figure 1.10 (a).

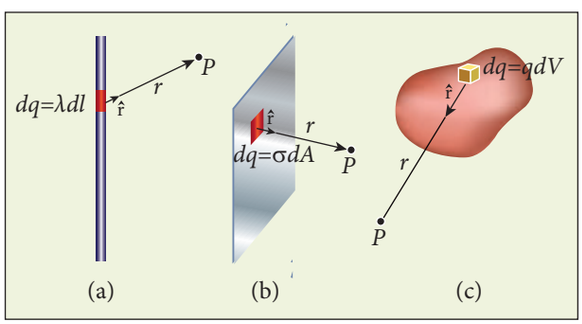


Figure 1.10 Line, surface and volume charge distribution

The electric field due to the line of total charge Q is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\lambda dl}{r^2}\hat{r} = \frac{\lambda}{4\pi\varepsilon_0}\int \frac{dl}{r^2}\hat{r}$$

(b) If the charge Q is uniformly distributed on a surface of area A, then surface charge density (charge per unit area) is $\sigma = \frac{Q}{A}$. Its unit is coulomb per square meter ($C\ m^{-2}$).

The charge present in the infinitesimal area dA is $dq = \sigma\, dA$. This is shown in the figure 1.10 (b).

The electric field due to a of total charge Q is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_\circ}\int \frac{\sigma da}{r^2}\hat{r} = \frac{1}{4\pi\varepsilon_\circ}\sigma\int \frac{da}{r^2}\hat{r}$$

This is shown in Figure 1.10(b).

(c) If the charge Q is uniformly distributed in a volume V, then volume charge density (charge per unit volume) is given by $\rho = \frac{Q}{V}$. Its unit is coulomb per cubic meter ($C\ m^{-3}$).

The charge present in the infinitesimal volume element dV is $dq = \rho dV$. This is shown in Figure 1.10(c).

The electric field due to a volume of total charge Q is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_\circ}\int \frac{\rho dV}{r^2}\hat{r} = \frac{1}{4\pi\varepsilon_\circ}\rho\int \frac{dV}{r^2}\hat{r}.$$

EXAMPLE 1.8

A block of mass m and positive charge q is placed on an insulated frictionless inclined plane as shown in the figure. A uniform electric field E is applied parallel to the inclined surface such that the block is at rest. Calculate the magnitude of the electric field E.

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